

Math 248 - HW 9

35. $\int_a^b f(x) dx = M(n) + (b-a) \frac{h^2}{24} f''(c)$ (midpoint)

a) $\int_a^b f(x) dx = T(n) - (b-a) \frac{h^2}{12} f''(c)$ (trapezoidal)

$$2 \int_a^b f(x) dx = M(n) + T(n) + (b-a) \frac{h^2}{24} f''(c) - (b-a) \frac{2h^2}{24} f''(c)$$

$$2 \int_a^b f(x) dx = M(n) + T(n) - \frac{h^2}{24}$$

$$\int_a^b f(x) dx = \frac{M(n) + T(n)}{2} - \frac{h^2}{24}$$

Trap Error $\rightarrow$
Midpoint Error $\rightarrow$
B.I. Error

Arbitrary $f(x) = x^3 + 4x^2 - 10$

True Value: $\int_0^{10} f(x) dx = 3733.33333333$

Midpoint: $\text{out} = \text{mid}(f, 0, 10, 100000)$
 $= 3733.333333175029$

Midpoint Error = $| \text{True} - \text{out} | = 1.5830391931 \times 10^{-7}$

Trapezoidal: $\text{out} = \text{trap}(f, 0, 10, 100000)$
 $= 3733.333333649982$

Trapezoidal Error = $| \text{True} - \text{out} | = 3.1664876587 \times 10^{-7}$

Better Integration: $\text{out} = \text{better_int}(f, 0, 10, 100000)$
 $= 3733.333333412297$

B.I. Error = $| \text{True} - \text{out} | = 7.8964148997 \times 10^{-8}$

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35. b) $\int_0^2 e^x dx = e^2 - 1 = 6.38905609893$ (True Val)

Midpoint:

$n = 100$

out = 6.388949615904636

error₁ = $1.06483026014 \times 10^{-4}$

$$\frac{\text{error}_1}{\text{error}_2} \approx 4$$

$n = 200$

out = 6.389029477941217

error₂ = $2.66209894333 \times 10^{-5}$

$$\frac{\text{error}_2}{\text{error}_3} \approx 4$$

$n = 400$

out = 6.389049443668735

error₃ = $6.65526191579 \times 10^{-6}$

Trapezoidal:

$n = 100$

out = 6.389269066047504

error₁ = $2.12967116854 \times 10^{-4}$

$$\frac{\text{error}_1}{\text{error}_2} \approx 4$$

$n = 200$

out = 6.389109340976075

error₂ = $5.32420454249 \times 10^{-5}$

$$\frac{\text{error}_2}{\text{error}_3} \approx 4$$

$n = 400$

out = 6.389069409458643

error₃ = $1.33105279927 \times 10^{-5}$

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35. c) $\int_0^2 e^x dx = e^2 - 1 = 6.38905609893$ (True Val)

Better Int:

$n = 100$

out = 6.389101007642736

error₁ = $4.4908712086 \times 10^{-5}$

$\frac{\text{error}_1}{\text{error}_2} \approx 4$

$n = 200$

out = 6.389067326125312

error₂ = $1.1227194662 \times 10^{-5}$

$\frac{\text{error}_2}{\text{error}_3} \approx 4$

$n = 400$

out = 6.389058905730356

error₃ = $2.80679970555 \times 10^{-6}$

Ratio of errors is the same as midpoint and trapezoidal approximations (~4) but on average the integral approximations were more accurate.

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36.

a)

$$\int_0^1 \sqrt{x} e^x dx \approx 1.255630082551864 \text{ (True Val)}$$

Midpoint:

$$n = 100$$

$$\text{out} = 1.255674146223198$$

$$\text{error}_1 = 4.40636713341 \times 10^{-5}$$

$$\frac{\text{error}_1}{\text{error}_2} \approx 2.546$$

$$n = 200$$

$$\text{out} = 1.255647391656979$$

$$\text{error}_2 = 1.73091051152 \times 10^{-5}$$

$$n = 400$$

$$\text{out} = 1.255636636927137$$

$$\text{error}_3 = 6.55437527319 \times 10^{-6}$$

$$\frac{\text{error}_2}{\text{error}_3} \approx 2.641$$

The ratio of errors is not 4 as expected, it is closer to 2.5

b) Since the midpoint approximation needs the second derivative f'' and since f'' is not bounded at $x=0$, the function is not well-behaved and the error blows up as $x \rightarrow 0$.

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36. c) $\int_0^1 \sqrt{x} e^x dx \rightarrow \int_0^1 2u^2 e^{u^2} du$

$$u = \sqrt{x}$$

$$x = u^2$$

$$dx = 2u du$$

Midpoint:

$$n = 100$$

$$\text{out} = 1.255539477650041$$

$$\text{error}_1 = 9.06049018228 \times 10^{-5}$$

$$n = 200$$

$$\text{out} = 1.255607430484084$$

$$\text{error}_2 = 2.26520677797 \times 10^{-5}$$

$$\frac{\text{error}_1}{\text{error}_2} \approx 4$$

$$n = 400$$

$$\text{out} = 1.255624419482271$$

$$\text{error}_3 = 5.66306959282 \times 10^{-6}$$

$$\frac{\text{error}_2}{\text{error}_3} \approx 4$$

It's working well this time because the second derivative of the new integrand $2u^2 e^{u^2}$ is well-behaved on $[0, 1]$, so the error doesn't blow up as $x \rightarrow 0$.